

Intelligent (Rational) Agents

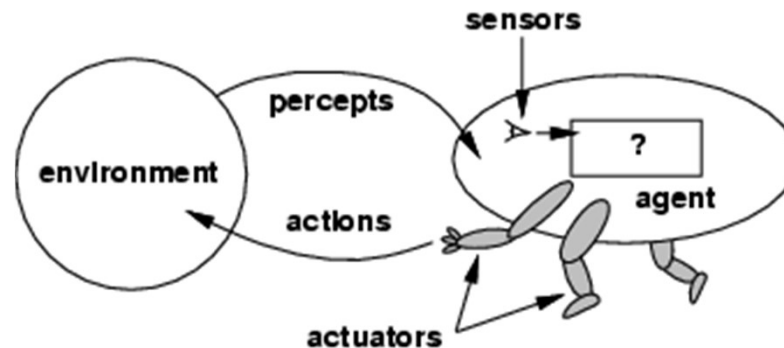
Outline

- Agents and environments
- Rationality
- PEAS (Performance measure, Environment, Actuators, Sensors)
- Environment types
- Agent types

Agents

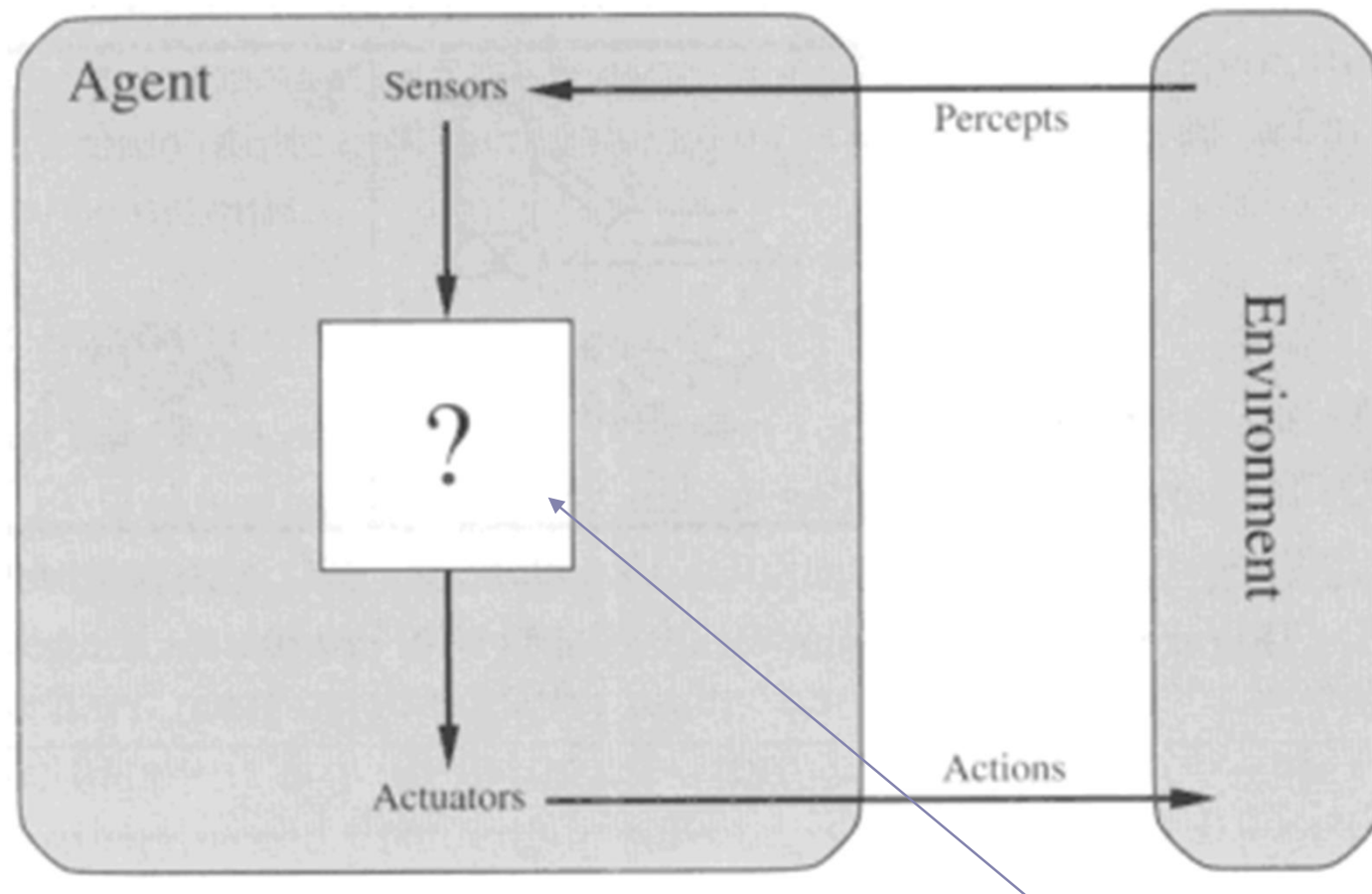
- An **agent** is anything that can be viewed as **perceiving** its **environment** through **sensors** and **acting** upon that environment through **actuators**
- Human agent:
 - eyes, ears, and other organs for sensors;
 - hands, legs, mouth, and other body parts for actuators
- Robotic agent:
 - cameras and infrared rangefinders for sensors
 - various motors for actuators
- A thermostat detecting room temperature.

Agents and environments



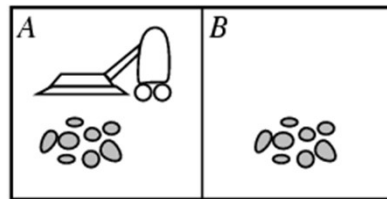
- The **agent function** maps from percept histories to actions:
$$[f: \mathcal{P}^* \rightarrow \mathcal{A}]$$
- The **agent program** runs on the physical **architecture** to produce f
- agent = architecture + program

Diagram of an agent



What AI should fill

Vacuum-cleaner world



- Percepts: location and contents, e.g., [A,Dirty]
- Actions: *Left, Right, Suck Up Dust, No Operation*
- *Agent's function* → *look-up table*
 - *For many agents this is a very large table*

Percept sequence	Action
[A, Clean]	<i>Right</i>
[A, Dirty]	<i>Suck</i>
[B, Clean]	<i>Left</i>
[B, Dirty]	<i>Suck</i>
[A, Clean], [A, Clean]	<i>Right</i>
[A, Clean], [A, Dirty]	<i>Suck</i>
⋮	⋮

Program implements the above agent function

Function Robotic-Vacuum-Agent(*[location,status]*) return an action

If *status = Dirty* **then return** *Suck*

else if *location = A* **then return** *Right*

else if *location = B* **then return** *left*

Rational agents

- **Rationality**
 - Performance measuring success
 - Agents prior knowledge of environment
 - Actions that agent can perform
 - Agent's percept sequence to date
- **Rational Agent:** For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

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PEAS

- A task environment specification includes the performance measure, the external environment, the actuators, and the sensors. In designing an agent, the first step must always be to specify the task environment as fully as possible.
- PEAS: Performance measure, Environment, Actuators, Sensors
- Must first specify the setting for intelligent agent design.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

Figure 2.5 Examples of agent types and their PEAS descriptions.

Homework

For each of the following Agent's, give a PEAS description of the task environment:

- Playing soccer.
- Exploring the subsurface oceans of Titan.
- Shopping for used AI books on the Internet.
- Playing a tennis match.
- Practicing tennis against a wall.
- Performing a high jump.
- Knitting a sweater.
- Bidding on an item at an auction.